

Biostatistics

#6_Multivariate analysis
PCA, FA, Cluster Analysis and LCA

Multivariate analysis

Multivariate analysis:

An analysis of a large number of variables or a group of variables consisting of a large number of variables.

Purpose of using this analysis

1. Simplify the structure (dimension contraction)
2. Exploring latent structures
3. Ordinate (give coordinates)
4. Numerical classification (clustering)

Course Content

This course will be divided in 11 chapters as follow

Fundamental series

- 1 Introduction
- 2 Group Comparison
- 3 General linear model (1) linear regression
- 4 Generalized linear model Logistic, Poisson
- 5 Survival analysis
- 6 **Principal component analysis, Factor analysis, Cluster analysis, Latent class analysis**

Advanced series Ref. Circulation CVD stat guideline

- 7 General linear model (2) ANOVA MIXED
- 8 Reporting Guide: PRISMA Meta-analysis
- 9 Reporting Guide: CONSORT/STROBE
- 10 Missing, Correlated data
- 11 Introduction to causal inference: DAG, propensity score

Analysis objectives and methods

- Dimensional contraction (one group) and ordination
 - PRINCOMP Principal component analysis (PCA)
 - FACTOR Factor Analysis
 - CORRESP Correspondence analysis Hayashi's quantification methods
 - MDS Multi Dimensional Scaling (MDS)
- Exploring latent structures
 - FACTOR Factor Analysis
 - CALIS Structure Equation Model
 - IRT Item response theory
 - LCA Latent class Analysis
 - LTA Latent transition analysis
- Dimensional contraction (two groups) and ordination
 - CANCORR Canonical correlation analysis
 - PLS Partial Least Squares
 - PLS Partial Least Squares
 - PCR Principal component regression
 - RRR Reduced Rank Regression

Analysis objectives and methods(cont.)

- **Supervised classification (discriminant analysis)**
 - DISCRIM Linear and quadratic discriminant analysis assuming normality
 - CANDISC canonical discriminant analysis
- **Unsupervised classification (clustering)**
 - VARCLUS Clustering of variables
Divides a set of numeric variables into disjoint or hierarchical clusters
 - FASTCLUS Clustering of individuals using the k-means method
Disjoint cluster analysis on the basis of distances computed from one or more quantitative variables
 - MODECLUS Clustering by Nonparametric Density Estimation
 - CLUSTER Distance-based Hierarchical Clustering
single linkage, complete linkage, Average linkage, Ward's, centroid, density
 - ACECLUS Obtains approximate estimates of the pooled within-cluster covariance matrix when the clusters are assumed to be multivariate normal with equal covariance matrices
 - TREE Produces a tree diagram, also known as a dendrogram or phenogram

Principal Components analysis and Factor Analysis

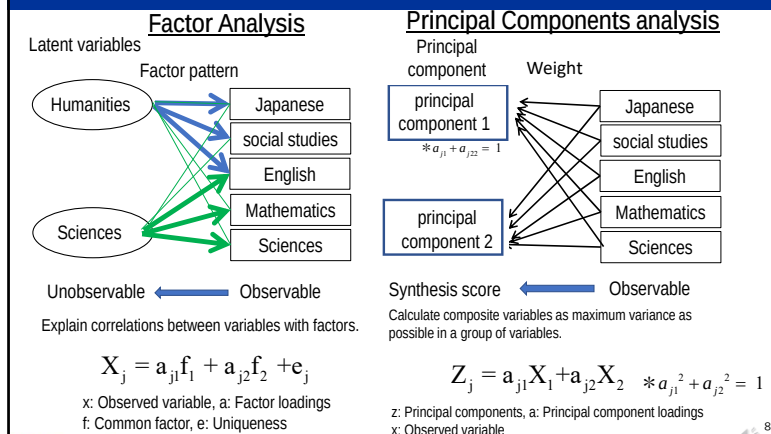
- **Principal Components analysis**
 - The components are calculated as linear combinations of the original variables
 - The goal is to explain **as much of the total variance in the variables as possible**
 - Use principal components analysis **to reduce the data into a smaller number of components.**
 - Maximize variance by orthogonal transformation of variable x
 - One orthogonal transformation after another
 - Eigenvalue decomposition of variance-covariance matrix (correlation matrix)
- **Factor Analysis**
 - The original variables are defined as linear combinations of the factors.
 - The goal in factor analysis is **to explain the covariances or correlations between the variables.**
 - Use factor analysis **to understand what constructs underlie the data.**
 - Rotate(orthogonal, oblique) so that the coefficients of the general factor are separated 0-1

Today's lecture

Today, We would like to follow the methods.

1. Principal Components analysis
2. Factor Analysis
3. Correspondence Analysis
4. Cluster analysis
5. Latent class analysis

Principal Components analysis and Factor Analysis



In Epidemiological Study Examples of PCA , FA Usage Dietary patterns and risk for type2 diabetes mellitus in UA Men

- Based on the dietary survey, dietary patterns were extracted by factor analysis.
- "Prudent Pattern", "Western pattern", etc. are extracted, and the incidence risk and other factors are examined using the dietary pattern scores.

1986		
	Prudent Pattern	Western Pattern
Vegetables†	0.75	—
Dark-yellow vegetables	0.64	—
Cruciferous vegetables	0.63	—
Green leafy vegetables	0.63	—
Legumes	0.63	—
Fruit	0.60	—
Tomatoes	0.54	—
Fish	0.51	—
Whole grains	0.38	—
Poultry	0.36	—
Salad dressing†	0.32	—
Fruit juice	0.32	—
Low-fat dairy products	0.16	—
Wine	—	—
Low-sugar drinks	—	—
Red meat	—	0.64
Processed meat	—	0.61
Refined grains	—	0.48
French fries	—	0.48
High-fat dairy products	—	0.47
Sweets and desserts	—	0.42
Eggs	—	0.42
Condiments	0.16	0.41
High-sugar drinks	—	0.36
Alcohol	—	0.26

. Dam RM, et al. Ann Intern Med. 2002;136:201-209.

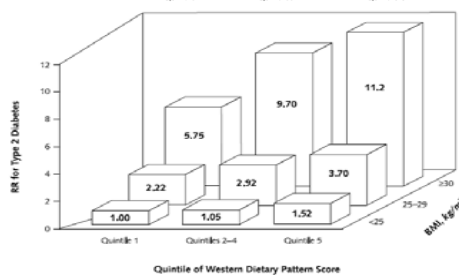
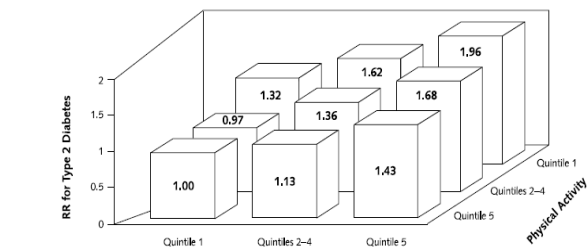
Principal Components analysis

Reduce the Dimensionality of Data

- The purpose of principal component analysis is to derive a small number of independent linear combinations (*principal components*) of a set of measured variables that capture as much of the variability in the original variables as possible.
- Principal component analysis is a dimension-reduction technique, as well as an exploratory data analysis tool.

Algorithm for Principal Component Analysis

- Calculate the center of gravity (mean value) of all data.
- From the center of gravity, calculate the direction in which the variance of the data is greatest (first principal component).
- Calculate where the variance is greatest in the direction that intersects the first principal component at right angles (orthogonal) (second principal component).
- Calculate the location where the variance is greatest in the direction perpendicular to the most recent principal component (third principal component).
- Repeat step ④ for each dimension of the data.



Western dietary pattern score and diabetes

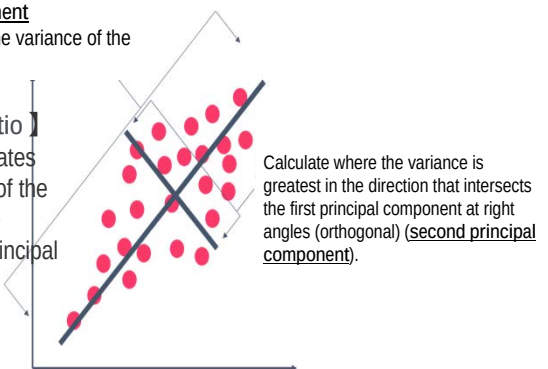
. Dam RM, et al. Ann Intern Med. 2002;136:201-209.

Principal Components analysis

first principal component

the direction in which the variance of the data is greatest.

【contribution ratio】
a number that indicates how many percent of the original data can be explained by this principal component alone



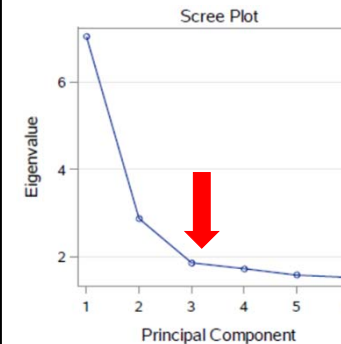
PCA Data & SAS code

Dataset (bdhq_1)
Dietary survey (FFQ) data for college students.

- Food Data
- Nutrient data

```
proc princomp data=work plots=all n=4 ;
var milk_p--yushi_p;
run;
```

The scree plot



• The scree plot graphs the eigenvalue against the component number.

• From the third component on, you can see that the line is almost flat, meaning the each successive component is accounting for smaller and smaller amounts of the total variance.

PROC princomp

• Eigenvalue

• This column contains the eigenvalues. The first component will always account for the most variance (and hence have the highest eigenvalue), and the next component will account for as much of the left over variance as it can, and so on.

• Cumulative

• This column sums up to proportion column, so that you can see how much variance is accounted for by, say, the first five components, .8800.

Eigenvalues of the Correlation Matrix				
	Eigenvalue	Difference	Proportion	Cumulative
1	7.03746311	4.17029914	0.2815	0.2815
2	2.86716397	1.01402361	0.1147	0.3962
3	1.85314036	0.13372938	0.0741	0.4703
4	1.71941098	0.14837288	0.0688	0.5391
5	1.57103810	0.05173205	0.0628	0.6019
6	1.51930605		0.0608	0.6627

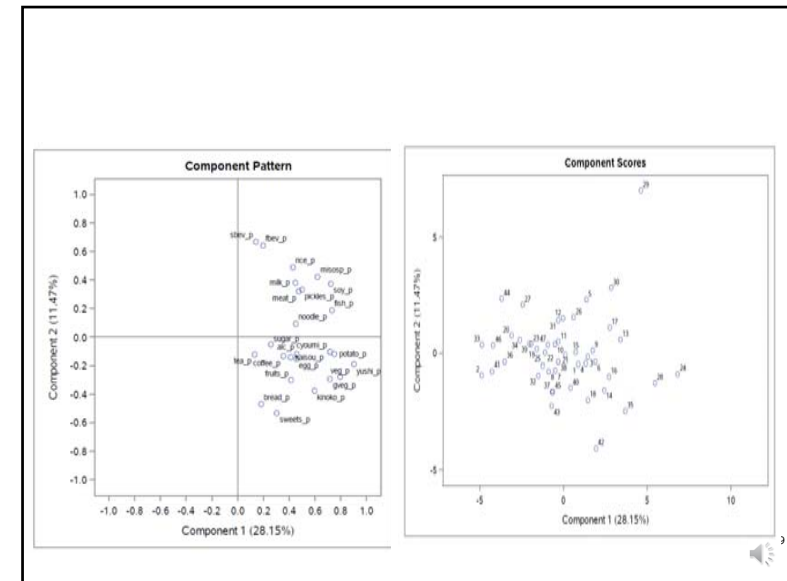
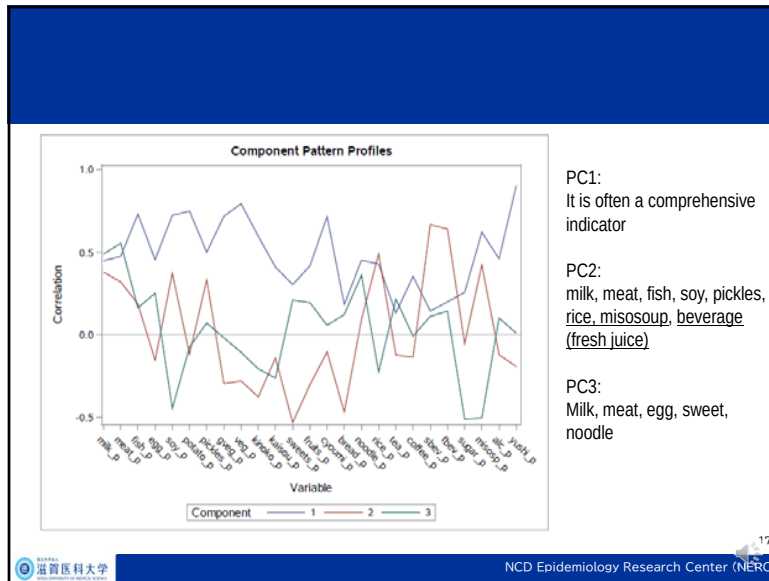
Eigenvector

Eigenvectors			
	Prin1	Prin2	Prin3
milk_p	0.160589	0.223718	0.358755
meat_p	0.179021	0.189031	0.407416
fish_p	0.275251	0.109312	0.118951
egg_p	0.171185	-0.091285	0.184937
soy_p	0.272396	0.220052	-0.209600
potato_p	0.282332	-0.094370	-0.054255
pickles_p	0.188073	0.196775	0.052696
grveg_p	0.270276	-0.174873	-0.013860
veg_p	0.299253	-0.165551	-0.078238
kinoko_p	0.234568	-0.220482	-0.151470
kaisou_p	0.154219	-0.028950	-0.193041
sweets_p	0.114510	-0.113886	0.154039
fruits_p	0.156808	-0.179271	0.142474
cyuomi_p	0.269580	-0.060919	0.042669
bread_p	0.068836	-0.764878	0.089913
noodle_p	0.169988	0.054439	0.285018
rice_p	0.161125	0.289137	-0.165405
tea_p	0.049317	-0.072210	0.158146
coffee_p	0.133531	-0.09532	-0.009523
sbev_p	0.053871	0.393539	0.081922
flwr_p	0.074814	0.177998	0.104394
sugar_p	0.096715	-0.01351	-0.375451
miscap_p	0.233803	0.248495	-0.369951
alc_p	0.173586	-0.071191	0.074977

• An eigenvector is a linear combination of the original variables. The components that have been extracted are orthogonal to one another, and they can be thought of as weights.

• These weights are multiplied by each value in the original variable, and those values are then summed up to yield the eigenvector.

• The strength of relationship between the variables and the components.



Score for students

Scores for each student

Obs	ID	Factor1	Factor2	Factor3	Factor4
1	1	0.52848	-0.33454	0.52161	1.61061
2	2	-0.57375	0.88566	-1.75360	-0.28321
3	3	-0.29932	-0.64578	0.38164	0.48721
4	4	-0.37014	1.66561	-0.10848	0.26260
5	5	0.48552	0.19543	0.23197	-1.04217
6	6	0.07094	-0.07745	0.08728	1.21810
7	7	-0.36687	-1.27952	-0.23341	0.06993
8	8	-0.29559	-0.78533	-0.44241	-0.30000
9	9	0.24201	-0.94228	1.16216	-0.33429
10	10	-0.05969	-0.52002	0.13275	0.07658
11	11	-0.03993	0.87889	0.05307	0.08199
12	12	0.85695	1.22394	-0.26620	-0.39812
13	13	-0.24025	0.39999	1.17623	2.16143
14	14	-0.39714	-0.55259	-0.25177	0.38090
15	15	-0.34589	0.37150	-0.17164	-0.58808
16	16	-0.17646	0.29287	0.69995	1.40661

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Factor analysis

- Factor analysis seeks to describe observed variables in terms of a smaller number of (unobservable) latent variables, or factors.
- A goal of factor analysis is to find a meaningful interpretation of the observed variables in terms of the unobserved factors. Another goal is to reduce the number of variables.

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Algorithm for Factor Analysis

- ① Calculate the "eigenvalues".
- ② Calculate factor loadings
"factor loadings" that indicate the strength of the influence of the common factors.
 - 【Commonality】
 - This is a numerical value that indicates the degree to which each observed variable can be explained by a group of factors (factor 1 and factor 2 in the table). It is a value between 0 (completely unexplainable) and 1 (completely explainable).
 - 1 - commonality = the amount of the unique factor.
 - 【factor contribution】
 - By dividing the factor contribution by the total number of observed variables, we can see how much the factor contributes (influences) to the total.
- ③ Rotating the factor axis
When the factor loadings of each observed variable are plotted on a scatter plot graph, it is often difficult to understand what the common factor refers to as it is, so to make it easier to interpret, we rotate the axes of the graph so that the numerical values of each factor are located along the axes.
- ④ Decide on the names of the common factors.
- ⑤ Calculation of factor scores

Calculation methods for factor extraction

1. Principal Factor Method (**method=prinit**)
The solution is obtained so that the factor contribution of the first factor is the largest.
2. Principal Component Method (**method=principal**) *default
 - Find the solution so that the factor contribution of each factor is as equal as possible.
 - The result of the principal component method without rotation is the same as the result of principal component analysis.
3. Maximum likelihood method (**method=ml**)
 - Estimates the solution by probability density. It is often used in covariance structure analysis.
 - It is said to provide accurate estimates even for data with distorted distributions.

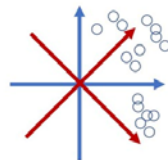
Rotate

- Rotate the estimated factor loadings matrix to get a simple structure.

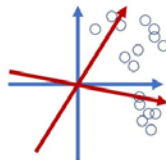
Orthogonal rotation

- The correlation between each factor is zero.
- Usually, a method called varimax rotation is used.

Varimax rotate



Promax rotate



Oblique rotation

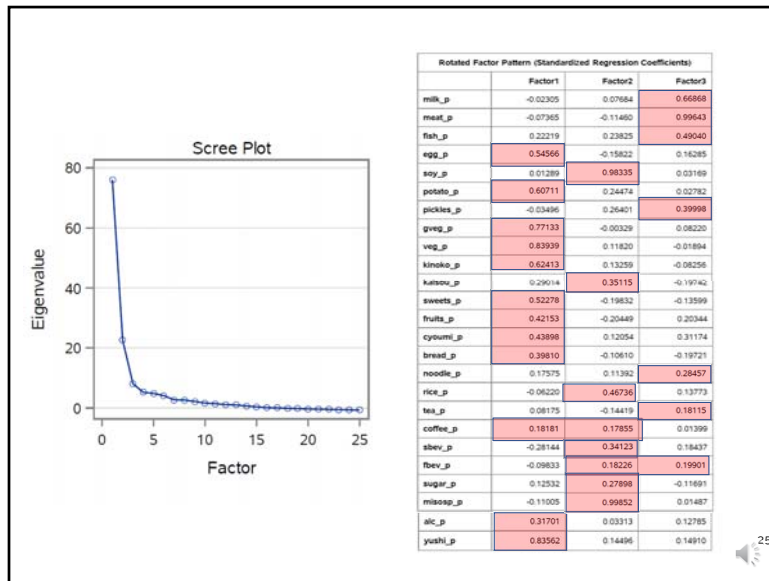
- There is a correlation between each factor.
- Usually, a method called promax rotation is used.

Factor Analysis (PROC factor)

```
proc factor data=work method=ml nfactor=4
  heywood rotate=promax plots=all out=score;
  var milk_p--yushi_p;
run;
```

Scores can be output as a data set for analysis using scores for each subject.

*heywood: Options for cases where the commonality estimate is greater than 1 (heywood case)



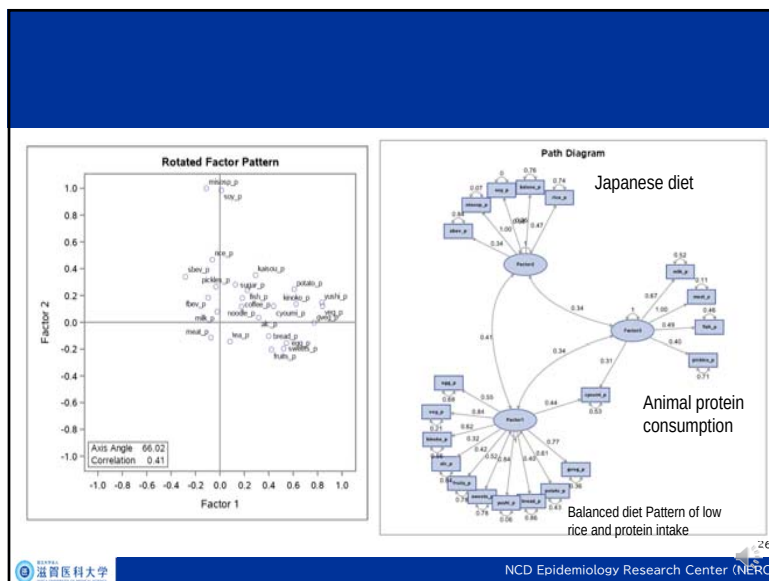
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Correspondence Analysis

Correspondence analysis is a useful data science visualization technique for finding out and displaying the relationship between categories.

- Step 1: Compute row and column averages
- Step 2: Compute the *expected values*
- Step 3: Compute the *residuals*
- Step 4: Plotting labels with similar residuals close together
- Step 5: Interpreting the relationship between row and column labels

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Example

Market survey on the impressions of the products

- The results of a market survey on the impressions of the products of companies A to G. The following nine items were tabulated with the answers.

- ① Easy to use
- ② Good design
- ③ Good design
- ④ Price is low
- ⑤ Trustworthy
- ⑥ Good color
- ⑦ Health-conscious
- ⑧ The material is good.
- ⑨ High quality

Contingency Table										
	use	design	buy	price	trust	color	healthy	Material	Premium	Sum
A	86	65	72	55	82	86	90	15	39	590
B	92	74	55	39	90	44	35	55	39	520
C	62	68	69	67	74	20	80	95	89	624
D	71	49	77	73	66	30	80	62	64	572
E	85	62	82	73	78	69	70	25	39	580
F	77	80	55	69	70	73	66	78	69	630
G	59	82	92	84	66	67	74	92	74	690
Sum	532	480	502	460	526	389	495	422	413	4210

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Correspondence Analysis PROC corresp

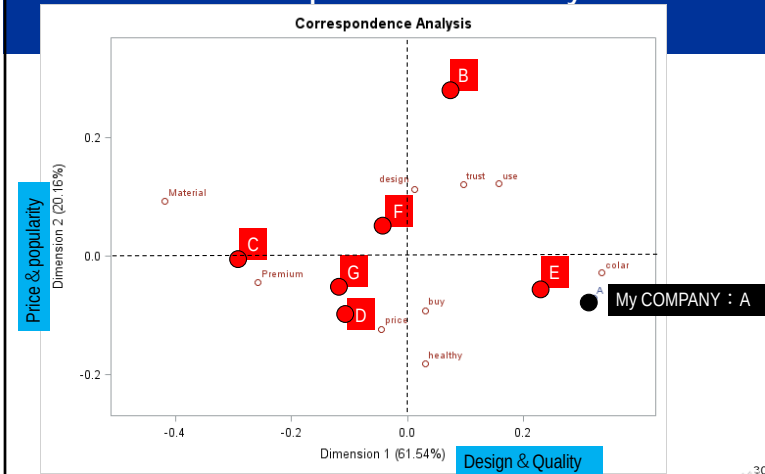
```
proc corresp all data=market dims=2 observed short
out=corpp;
var use design buy price trust color healthy Material Premium;
id company;
run;
```

```
proc sgplot data = corpp noautolegend;
axis min = -.4 max = .4 values=(-.3 to .3 by .1) valueshint;
axis min = -.3 max = .3;
scatter x = dim1 y = dim2 /group = _type_
MARKERCHAR = company
markercharattrs=(size=10 weight=bold);
run;
```

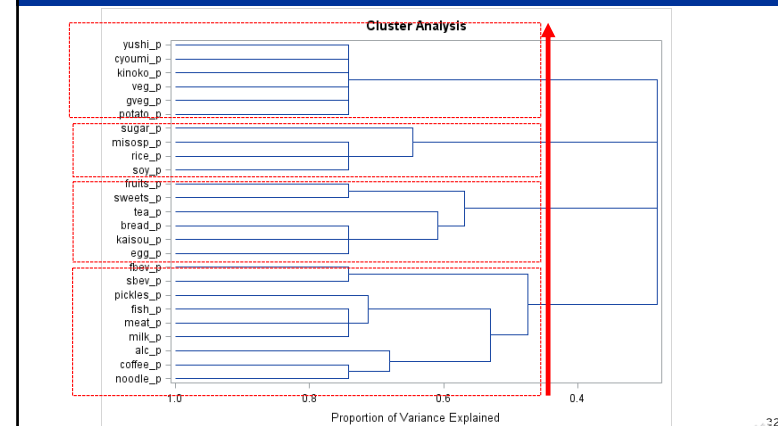
Cluster analysis

- Unsupervised classification (clustering)
 - VARCLUS Clustering of variables
Divides a set of numeric variables into disjoint or hierarchical clusters
 - FASTCLUS Clustering of individuals using the k-means method
Disjoint cluster analysis on the basis of distances computed from one or more quantitative variables
 - MODECLUS Clustering by Nonparametric Density Estimation
 - CLUSTER Distance-based Hierarchical Clustering
single linkage, complete linkage, average linkage, Ward's, centroid, density
 - ACECLUS Obtains approximate estimates of the pooled within-cluster covariance matrix when the clusters are assumed to be multivariate normal with equal covariance matrices
 - TREE Produces a tree diagram, also known as a dendrogram or phenogram

Correspondence Analysis

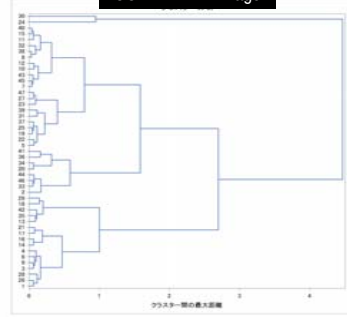


VARCLUS Clustering of variables



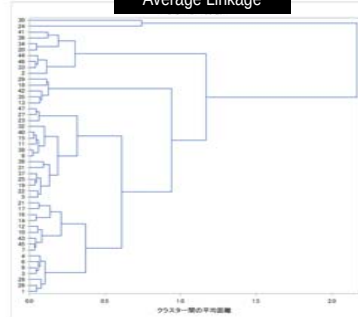
PROC CLUSTER

COMPLETE Linkage



the distance between two clusters is the maximum distance between an observation in one cluster and an observation in the other cluster

Average Linkage



the distance between two clusters is the average distance between pairs of observations, one in each cluster

Optional (Latent Class Analysis)

Report

- For the following Q1-Q3 assignments shown in the lecture slides, check that the results can be reproduced using a statistical package such as SAS or SPSS, and submit the output.

- Principal Components analysis
- Factor Analysis
- Correspondence Analysis
- Cluster analysis
- Latent class analysis

Use for Food Frequency Questionnaire data
Survey data for products

* Review Q4 and 5 if you have time.
LCA is supported by SAS as a macro, SPSS is not supported as a package.

Please submit via web class

Latent class analysis (LCA) How it differs from factor analysis

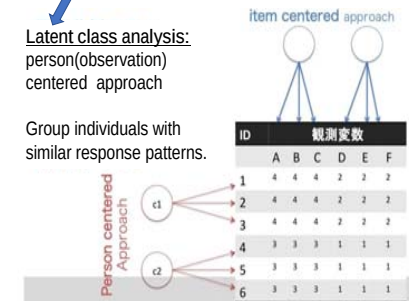
- Latent class analysis enables to find clusters of observations for categorical response variables. A latent variable is an unobservable grouping variable.

Factor analysis: Item centered approach
Summarize variables with similar response patterns

Latent class analysis:
person(observation) centered approach

- Each level of the latent variable is called a latent class.

- The Latent Class Analysis platform fits a latent class model and determines the most likely cluster or latent class for each observation.



Example

- Data from a 9-item survey of 1,000 people on Alcohol drinking
- coded 1 for yes and 0 for no.
- Load survey dataset 'Lcadat.csv'

item1 = "I like to drink"
 item2 = "I drink hard liquor"
 item3 = "I have drank in the morning"
 item4 = "I have drank at work"
 item5 = "I drink to get drunk"
 item6 = "I like the taste of alcohol"
 item7 = "I drink help me sleep"
 item8 = "Drinking interferes with my relationships"
 item9 = "I frequently visit bars";

A selection of n=10 was checked.

id	item1	item2	item3	item4	item5	item6	item7	item8	item9
1	1	0	0	0	0	0	0	0	0
2	1	1	0	1	1	1	1	0	0
3	1	0	0	0	0	0	0	0	0
4	1	0	0	0	0	1	1	0	0
5	1	0	0	0	1	0	0	0	1
6	0	1	0	0	0	1	0	0	0
7	1	1	0	0	0	0	0	0	1
8	1	0	1	0	0	0	0	0	0
9	1	0	0	0	0	0	0	1	0
10	0	0	0	0	0	1	0	0	0

Example

```

data lca1;
  infile "F:\lacadat.csv" delimiter=",";
  input id item1 item2 item3 item4 item5
        item6 item7 item8 item9;
  label item1 = "I like to drink"
        item2 = "I drink hard liquor"
        item3 = "I have drank in the morning"
        item4 = "I have drank at work"
        item5 = "I drink to get drunk"
        item6 = "I like the taste of alcohol"
        item7 = "I drink help me sleep"
        item8 = "Drinking interferes with my
relationships"
        item9 = "I frequently visit bars";

```

```
run;
```

```

options label nocenter nodate;
proc print data = lca1 (obs=10)
noobs;
run;
proc means data = lca1 mean std;
var item1 - item9;
run;

```

Analysis of LCA

```

data lca2;
  set lca;
  array it(9) item1 - item9;
  do i = 1 to 9;
    it(i) = 2 - it(i);
  end;
  drop i;
Run;

proc lca data = lca2 OUTPARAM = test
outpost = lca1_post;
nclass 3;
id id;
items item1 item2 item3 item4 item5
item6 item7 item8 item9;
categories 2 2 2 2 2 2 2 2 2 ;
seed 123741;
run;

```

The current version of the LCA program does not support 0/1 variables, so in the DATA step on the left, the item needs to be set to 1/2.

The option outparam allows us to save the final parameter estimates for each item and each class and outpost allows us to save the posterior probabilities for each observation.

```

data test2;
  set test;
  itemid = substr(variable, 5,2);
  if _n_ = 1 then do;
    call symput('p1', put(estlc1, percent10.2));
    call symput('p2', put(estlc2, percent10.2));
    call symput('p3', put(estlc3, percent10.2));
  end;
run;

```

```

goptions reset = all;
symbol1 i=join v = triangle c=green;
symbol3 i=join v = circle c=red;
symbol2 i=join v = square c=blue;
axis1 label=(a=90 "Probability");
axis2 label=("items");
legend down=3 label=none value=(font=swiss "class 1 &p1" "class 2 &p2" "class 3 &p3")
  position=(top right inside) mode=share cborder=black;
proc gplot data = test2;
  where respcat =1;
  plot(estlc1 estlc2 estlc3)*itemid /vaxis = axis1 legend=legend1 haxis=axis2 overlay;
run;
quit;

```

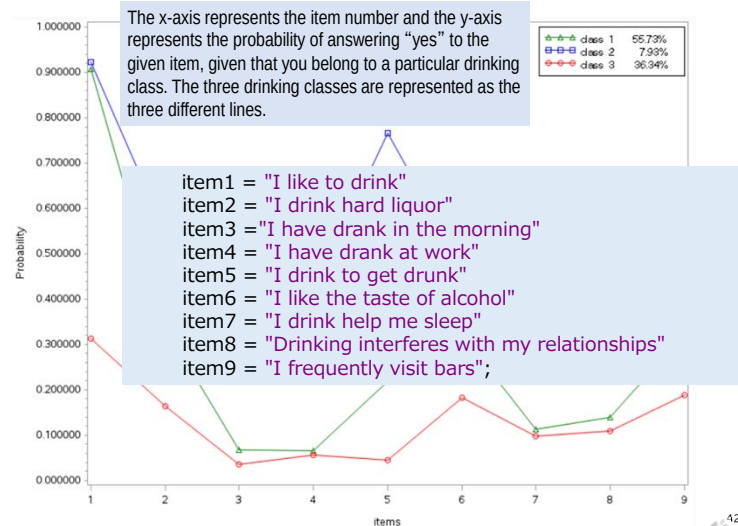


Class Membership

```
proc print data = lca1_post (obs = 10) noobs; run;
```

d	item1	item2	item3	item4	item5	item6	item7	item8	item9	POSTLC1	POSTLC2	POSTLC3	BEST
1	1	2	2	2	2	2	2	2	2	0.644384	0.001339	0.354277	1
2	1	1	2	1	1	1	1	2	2	0.098343	0.900471	0.001185	2
3	1	2	2	2	2	2	2	2	2	0.644384	0.001339	0.354277	1
4	1	2	2	2	2	1	1	2	2	0.796411	0.025913	0.177676	1
5	1	2	2	2	1	2	2	2	1	0.934042	0.025114	0.040844	1
6	2	1	2	2	2	1	2	2	2	0.311499	0.002387	0.686115	3
7	1	1	2	2	2	2	2	2	1	0.903132	0.004933	0.091934	1
8	1	1	2	2	2	2	2	2	1	0.765404	0.016531	0.218065	1
9	1	1	2	2	2	2	2	2	1	0.695444	0.014447	0.290109	1
10	1	1	2	2	2	2	2	2	1	0.148809	0.000483	0.850708	3

The outpost option on the PROC LCA statement creates an output file which contains the original data used in the analysis (i.e., item1 to item9) followed by the probability estimates that the observation belongs to Class 1, Class2, and Class 3. Next, the class with the highest probability (the modal class) is shown. The first 10 observations are shown this slide..



Cardiovascular Damage Phenotypes and All-Cause and CVD Mortality in Older Adults

- 5067 older adults from the Cardiovascular Health Study followed for up to 6 years.
- Using latent class analysis (LCA), we created CV damage phenotypes probabilities of
 - abnormal brain infarctions
 - major echocardiogram abnormalities
 - N-terminal probrain natriuretic peptide
 - troponin T
 - interleukin-6
 - c reactive-protein
 - galectin-3
 - cystatin C.
- Cox-proportional hazards regression to evaluate the association between the categorical CV damage phenotype and all-cause and CVD-mortality.

Miller LM, et al. Ann Epidemiol. 2021 Nov;63:35-40



NCD Epidemiology Research Center (NERC)



Cardiovascular Damage Phenotypes and All-Cause and CVD Mortality in Older Adults

Number of Claves and Suggested Phenotypes for 5 class model							Unadjusted	Adjusted ^a
	Class 1	Class 2	Class 3	Class 4	Class 5		HR (95% CI)	
	Cardio-Renal	Cardiac	Healthy	Multisystem	Inflammatory	Phenotype		
Membership Probability	0.11	0.18	0.53	0.07	0.11	Healthy (ref)	-	-
Cortical Brain Infarct	0.45	0.40	0.23	0.51	0.12	Cardio-Renal	3.30 (2.96, 3.67)***	2.31 (2.02, 2.64)***
ECG	0.46	0.73	0.20	0.89	0.13	Cardiac	2.39 (2.16, 2.65)***	1.76 (1.55, 1.99)***
NTproBNP	0.60	0.55	0.11	0.99	0.14	Multisystem Morbidity	5.59 (4.91, 6.37)***	4.02 (3.44, 4.70)***
Troponin T	0.91	0.81	0.33	0.98	0.45	Inflammatory	1.58 (1.38, 1.81)***	1.50 (1.28, 1.76)***
ST2	0.37	0.43	0.24	0.64	0.19	CVD-Mortality 1528 deaths		
CRP	0.11	0.05	0.05	0.55	0.52	Phenotype		
IL-6	0.41	0.26	0.13	0.99	0.98	Healthy (ref)	-	-
GaI3	0.69	0.15	0.16	0.56	0.33	Cardio-Renal	3.33 (2.85, 3.89)***	2.29 (1.89, 2.77)***
Cystatin C	0.96	0.10	0.04	0.58	0.18	Cardiac	3.22 (2.82, 3.68)***	2.22 (1.88, 2.62)***
						Multisystem Morbidity	6.80 (5.66, 8.17)***	4.90 (3.95, 6.06)***
						Inflammatory	1.58 (1.30, 1.91)***	1.46 (1.17, 1.81)**

**
P<0.001
P<0.01

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To perform an LCA

•SAS

For SAS, please download and apply the LCA macro program from the following site.

- Download proper version of PROC LCA
 - a. Go to <https://scholarsphere.psu.edu/resources/cddf9258-09f5-49fe-a313-4753e643b48ab>.
 - b. Sign in or register
 - c. Download the proper PROC, above (and users' guide)

•R

- Download proper version of OFFD sdfndjh
 - <https://scholarsphere.psu.edu/resources/8aa5cc3f-2d07-4c9d-9659-223aba0a4a8d>

•STATA

- Download proper version of OFD Vwdwd Sx:j lq
 - <https://scholarsphere.psu.edu/resources/c2e33d7c-133a-4b9e-a8ba-ed78044037e0>

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